

Lecture- 23

Mathematics 1 (15B11MA111)

**Module: Vector Differential
and Integral Calculus**

CO [C105.4]

Topics to be covered

- **Gauss Divergence Theorem**

- ❖ **Definition**

- ❖ **Examples**

- ❖ **Practice Problems**

- **Reference for the lecture**

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Divergence Theorem of Gauss

Definition: Let D be a closed and bounded region in the three dimensional space whose boundary is a piecewise smooth surface S that is oriented outward. Let

$v(x, y, z) = v_1(x, y, z)i + v_2(x, y, z)j + v_3(x, y, z)k$ be a vector field for which v_1, v_2, v_3 are continuous and have continuous first order partial derivatives in some domain containing D . Then

$$\iint_S v \cdot n \, dA = \iiint_D \nabla \cdot v \, dV = \iiint_D \operatorname{div}(v) \, dV$$

Where n is the outer unit normal vector to S .

Remark: In terms of components of $\mathbf{v}$, divergence theorem can be written as

$$\iint_S v_1 dy dz + v_2 dz dx + v_3 dx dy = \iiint_D \left(\frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z} \right) dx dy dz$$

Physical Interpretation of Divergence Theorem

Consider a small sphere S_r of radius r , centered at a fixed point $P_0(x_0, y_0, z_0)$. Let D_r be the spherical volume bounded by S_r . Then by divergence theorem, we obtain

$$\iint_S v \cdot n \, dA = \iiint_D \nabla \cdot v \, dV$$

Now, approximate $(\nabla \cdot v)(P(x, y, z)) \approx (\nabla \cdot v)P_0(x_0, y_0, z_0)$. where (x, y, z) is an arbitrary point in D_r .

$$\iint_S v \cdot n \, dA = (\nabla \cdot v)(P_0) \iiint_{D_r} dV = V_r (\nabla \cdot v)(P_0)$$

Where V_r is the volume of the small sphere. Let $r \rightarrow 0$, so that the sphere D_r tends to the point P_0 .

Then
$$(\nabla \cdot v)(P_0) = \lim_{r \rightarrow 0} \frac{1}{V_r} \iint_{S_r} (v \cdot n) dA$$

We conclude that the $\text{div}(v)$ is flux per unit volume.

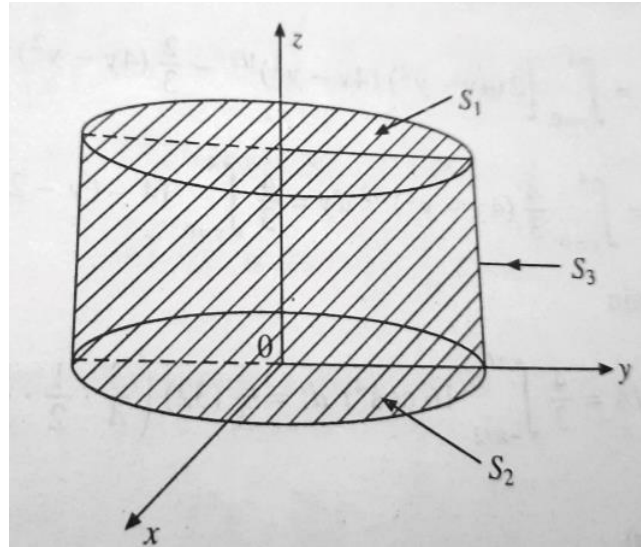
Example 1: Let d be the region bounded by the closed cylinder

$x^2 + y^2 = 16, z = 0$ and $z = 4$. Verify the divergence theorem

if $v = 3x^2i + 6y^2j + zk$

Solution: We have, $\nabla \cdot v = 6x + 12y + 1$. Therefore,

$$\iiint_D \nabla \cdot v dV = \int_{z=0}^4 \int_{x=-4}^4 \int_{y=-\sqrt{16-x^2}}^{\sqrt{16-x^2}} (6x + 12y + 1) dy dx dz$$



Since x, y are odd functions, we obtain

$$\iiint_D \nabla \cdot \mathbf{v} \, dV = 4.2.2 \int_{x=0}^4 \int_{y=0}^{\sqrt{16-x^2}} dy dx = 64\pi$$

The surface consists of three parts, S_1 (top), S_2 (bottom), S_3 (vertical)

On S_1 : $z = 0, n = k$

$$\begin{aligned}\iint_{S_1} v \cdot n \, dA &= \iint_{S_1} z \, dA = 4 \iint_{S_1} dA \\ &= 4 \text{ (area of the circular region with radius 4)} \\ &= 64\pi\end{aligned}$$

On S_2 : $z = 0, n = -k$

$$\iint_{S_2} v \cdot n \, dA = \iint_{S_2} -z \, dA = 0$$

On S_3 : $x^2 + y^2 = 16, n = \frac{2xi+2yj}{2\sqrt{x^2+y^2}} = \frac{1}{4}(xi + yj)$

$$\iint_{S_3} v \cdot n \, dA = \frac{1}{4} \iint_{S_2} (3x^3 + 6y^3) \, dA$$

Using cylindrical coordinates, $x = 4\cos\theta$, $y = 4\sin\theta$, $dA = 4d\theta dz$

Therefore

$$\iint_{S_3} v \cdot n \, dA = \int_{z=0}^4 \int_{\theta=0}^{2\pi} [192\cos^3\theta + 384\sin^3\theta] 4d\theta \, dz = 0$$

Hence,

$$\iint_S v \cdot n \, dA = \iiint_D \nabla \cdot v \, dV$$

Example 2: Find the flux of the vector field $\mathbf{v} = z \mathbf{i} + y \mathbf{j} + x \mathbf{k}$ over the unit sphere $x^2 + y^2 + z^2 = 1$.

Solution: First we compute the divergence of $\mathbf{v}$:

$$\operatorname{div} \mathbf{v} = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z} = 1$$

The unit sphere S is the boundary of the unit ball D given by $x^2 + y^2 + z^2 \leq 1$.

$$\text{flux} = \iint_S (\mathbf{v} \cdot \mathbf{n}) \, dA$$

Using divergence theorem

$$\iint_S v \cdot n \, dA = \iiint_D \nabla \cdot v \, dV = \iiint_D \operatorname{div}(v) \, dV$$

Therefore,

$$\text{flux} = \iint_S (v \cdot n) \, dA = \iiint_D \operatorname{div}(v) \, dV$$

$$= \iiint_D \operatorname{div}(v) \, dV = \iiint_D 1 \, dV$$

$$= V(D) = \frac{4\pi}{3}$$

Practice Problems

Q.1: Use Divergence theorem to evaluate $\iint_S \vec{F} \cdot d\vec{S}$, where $\vec{F} = 4x\vec{i} - 2y^2\vec{j} + z^2\vec{k}$

taken over the region bounded by the cylinder $x^2 + y^2 = 4$, $z = 3$.

Q.2: Verify Divergence theorem, where $\vec{F} = 4xz\vec{i} - y^2\vec{j} + yz\vec{k}$ and S is the surface of the cube bounded by the planes, $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$.

Thank You